Introduction to NoSQL Databases

MongoDB, Key-Value, Document and Graph Models

NoSQL versus RDBMS: Concepts, Use Cases and Comparison

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Class Objectives

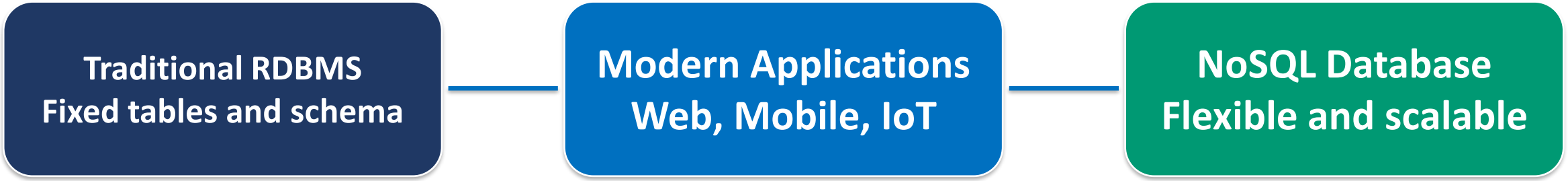
What you will learn in this class

Understand NoSQL | **Identify Models** | **Learn MongoDB** | **Compare RDBMS** | **Choose Database**

- Explain the need for NoSQL databases.
- Define NoSQL and its characteristics.
- Identify major NoSQL database models.
- Explain MongoDB as a document database.
- Describe key-value, document, and graph models.
- Compare NoSQL with RDBMS.
- Select a suitable database for an application.

Why Do We Need NoSQL?

From structured tables to flexible large-scale data



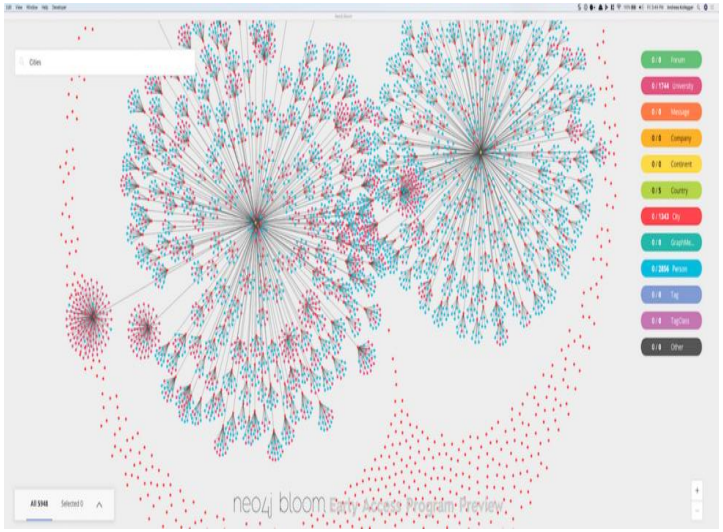
- Modern applications generate semi-structured and unstructured data.
- Examples: social-media posts, sensor data, product catalogues, logs, user profiles.
- Data structure may change frequently.
- Applications may require high availability and horizontal scalability.
- NoSQL supports flexible data models and distributed storage.

Major Types of NoSQL Databases

Different models for different requirements

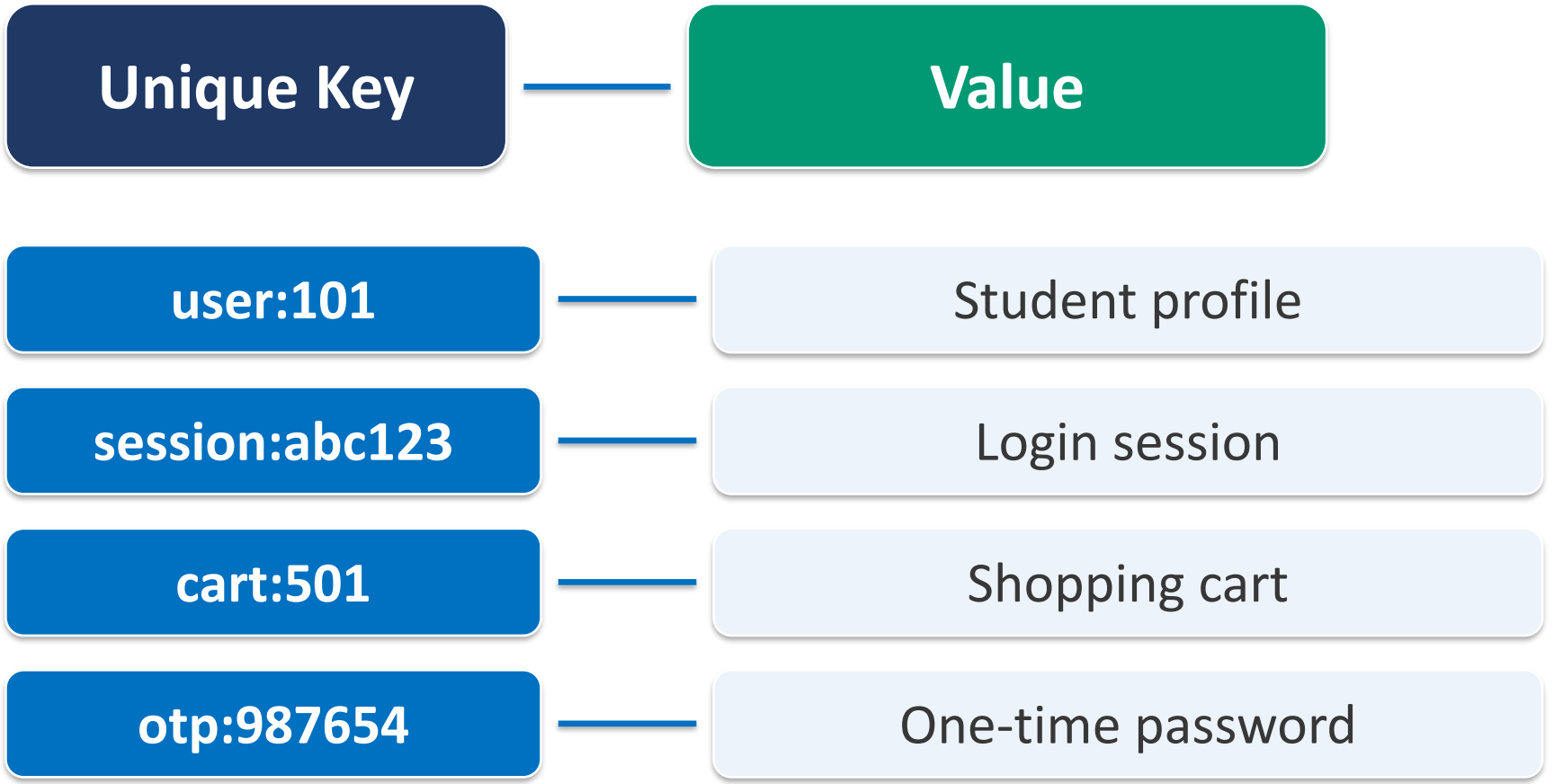
Key-Value	Document	Column-Family	Graph
Key → Value	JSON / BSON	Rows & Families	Nodes & Edges
Example: Redis	Example: MongoDB	Example: Cassandra	Example: Neo4j

- Key-value databases provide very fast access using a unique key.
- Document databases store flexible JSON-like records.
- Column-family databases support large distributed workloads.
- Graph databases focus on relationships between entities.
- Today's focus: MongoDB, key-value, document, and graph models.



1. Key-Value Database

Fast storage and retrieval using unique keys



- The key uniquely identifies the value.
- Retrieval is very fast when key is known.
- Values may be text, JSON, or binary data.
- Examples: Redis, DynamoDB, Riak.
- Best for caching, sessions, carts, OTPs.

Use Cases: Key-Value Stores

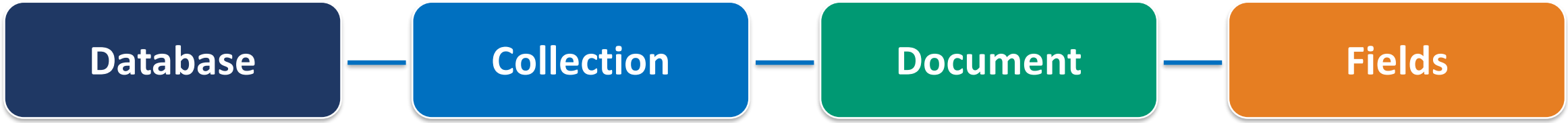
Fast access using a known key

Web Session	Caching	Shopping Cart
Login status and user session	Frequently accessed data	Temporary selected products

OTP / Token	Leaderboard	Rate Limiting
Temporary verification codes	Real-time scores and ranks	Controls API request rate

2. Document Database

Flexible storage of JSON-like records

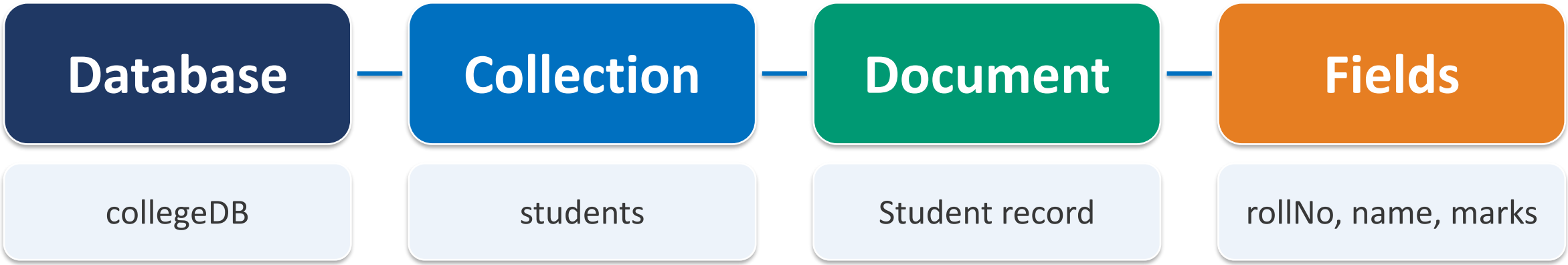


- A document contains field-value pairs.
- Fields can store strings, numbers, dates, arrays, and nested documents.
- Documents in one collection may have different fields.
- MongoDB stores documents in BSON, a binary form of JSON.
- A collection is similar to a table; a document is similar to a row.

```
{ "rollNo": 101, "name": "Anita", "branch": "CSE", "marks": 85,
"skills": ["Python", "SQL", "MongoDB"],
"address": { "city": "Hyderabad", "state": "Telangana" } }
```

MongoDB: Document Database

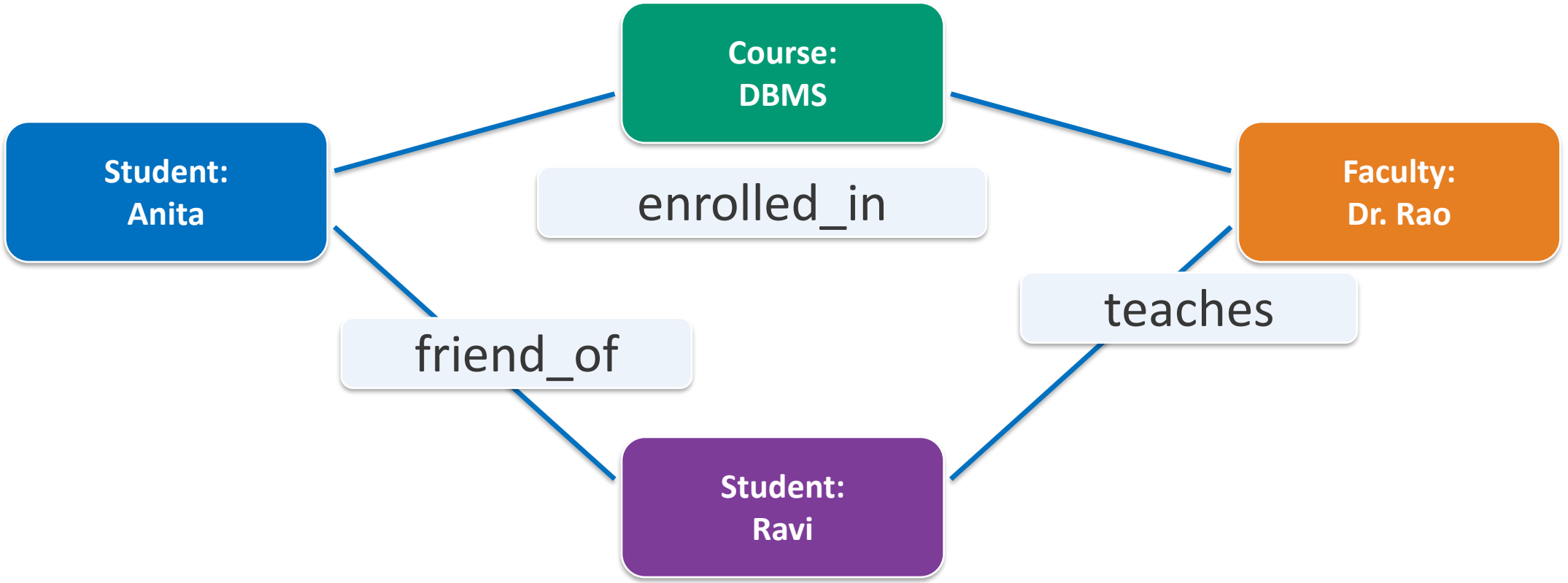
Database → Collection → Document → Field



- MongoDB is a document-oriented NoSQL database.
- It stores data in BSON, a binary representation of JSON.
- A collection contains related documents.
- Each document can have a different structure.
- Arrays and nested documents reduce the need for joins.

3. Graph Database

Data as nodes and relationships as edges



- Nodes represent entities: students, courses, faculty, products.
- Edges represent relationships: enrolled_in, teaches, friend_of.
- Graph databases are useful for highly connected data.
- Examples: Neo4j, Amazon Neptune, ArangoDB.
- Use cases: recommendations, fraud detection, social networks.

Comparison of NoSQL Models

Choose the model according to workload

Model	Data Unit	Main Strength	Suitable Use Cases	Example
Key-Value	Key and value	Very fast lookup using a key	Caching, sessions, carts	Redis
Document	JSON/BSON document	Flexible structure	Profiles, catalogs, IoT data	MongoDB
Graph	Nodes and edges	Relationship processing	Social networks, recommendations	Neo4j
Column-Family	Rows and column families	Large-scale distributed storage	Time-series and analytics	Cassandra

NoSQL versus RDBMS

Two complementary approaches

RDBMS
Tables, Rows, Columns

NoSQL
Flexible Data Models

Aspect	RDBMS	NoSQL
Schema	Fixed and predefined	Flexible or dynamic
Data model	Tables, rows, columns	Key-value, document, graph
Relationships	Foreign keys and joins	Embedding, references, graph edges
Scaling	Mostly vertical scaling	Often horizontal scaling
Consistency	Strong ACID consistency	Often BASE / eventual consistency
Best for	Structured transactional data	Flexible, large-scale, distributed data

Choosing the Right Database

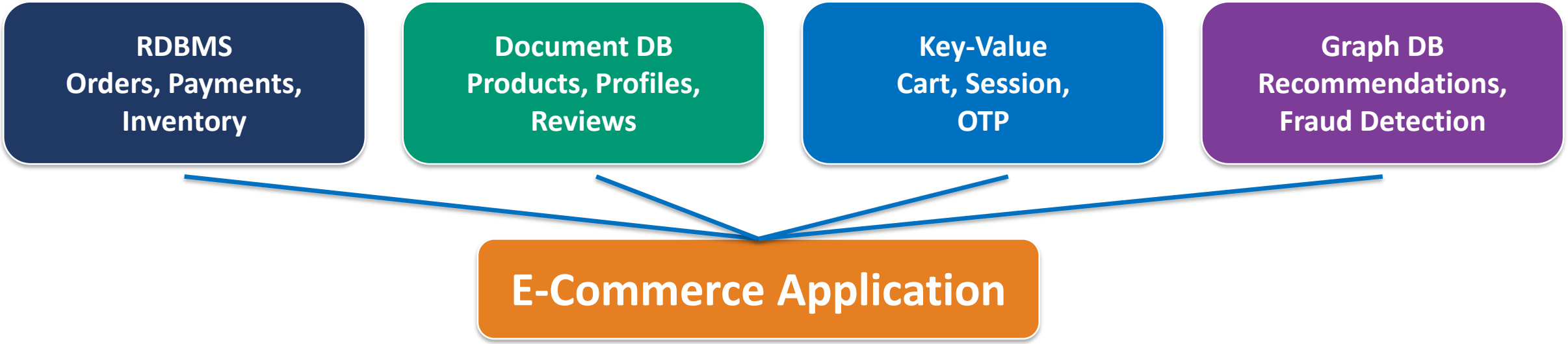
Use the right database for the right workload

RDBMS	Key-Value	Document	Graph
Structured data, ACID transactions	Fast lookup, caching, sessions	Flexible JSON-like data	Relationships and connected data

- Use RDBMS for banking, inventory, payroll, and academic records.
- Use key-value stores for sessions, carts, caching, and temporary tokens.
- Use document databases for product catalogues, profiles, and IoT data.
- Use graph databases for recommendations, social networks, and fraud detection.
- Modern systems often use multiple databases: polyglot persistence.

Example: E-Commerce Application

Using multiple databases for different needs



- RDBMS manages transactional and financial data.
- Document database stores flexible catalogue and profile data.
- Key-value store handles temporary and high-speed data.
- Graph database supports relationship-based recommendations.
- The correct database is selected based on the workload.

Classroom Activity

Identify the most suitable database model

Student login sessions	**Key-Value**
Product details with varying attributes	**Document**
Friends-of-friends in a social network	**Graph**
Bank transactions with strict consistency	**RDBMS**
IoT temperature readings	**Document / Column-Family**
Product recommendations	**Graph**

Exit Quiz

- What does NoSQL stand for?
- Name four major types of NoSQL databases.
- What is a document in MongoDB?
- How does MongoDB represent arrays and nested data?
- How does a graph database represent entities and relationships?
- Give one use case each for key-value, document, and graph databases.
- State any three differences between RDBMS and NoSQL.
- Which database model is most suitable for an e-commerce shopping cart? Justify your answer.

Summary

Key points to remember

NoSQL	Key-Value	Document	Graph	RDBMS
Flexible, scalable, non-relational	Fast access using unique keys	JSON-like records; MongoDB	Nodes and relationships	Structured data and ACID transactions

- NoSQL and RDBMS are complementary technologies.
- Use the database model that matches the data structure and workload.
- MongoDB is suitable for flexible document-oriented applications.
- Graph databases are ideal when relationships are central.
- Database selection should consider schema flexibility, scalability, consistency, and query needs.